

ANIAK, AK, USA

WMO: **702320**Lat: **61.582N**Lon: **159.543W**Elev: **26**StdP: **101.01**Time Zone: **-9.00 (NAK)**Period: **94-19**WBAN: **26516**

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-37.5	-33.4	-34.0	0.2	-28.2	-33.0	0.2	-26.9	10.8	1.5	9.2	-0.6	0.2	120	0.713

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.0	23.0	14.5	21.4	14.0	19.1	13.2	15.9	21.3	14.9	19.7	14.0	18.3	2.7	340

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
13.7	9.8	17.4	12.5	9.1	16.3	12.0	8.8	15.9	44.5	21.7	41.3	19.7	39.0	18.4	21.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.3	7.3	6.1	-38.6	27.0	4.8	2.4	-42.0	28.7	-44.8	30.0	-47.5	31.4	-51.0	33.1		
(5)			-34.1	17.6	3.0	1.7	-36.3	18.9	-38.0	19.9	-39.7	20.8	-41.9	22.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-0.3	-14.4	-9.9	-9.6	-0.5	7.2	12.3	13.8	12.0	7.7	-0.3	-9.0	-13.2
	DBStd	12.25	11.09	9.58	7.76	5.79	4.13	2.86	2.68	2.65	3.45	5.83	8.10	9.78
	HDD10.0	4044	755	556	606	315	103	9	2	10	82	319	569	718
	HDD18.3	6791	1013	789	864	565	345	182	142	197	320	576	819	977
	CDD10.0	298	0	0	0	0	17	78	120	71	12	1	0	0
	CDD18.3	4	0	0	0	0	0	1	2	1	0	0	0	0
	CDH23.3	68	0	0	0	0	2	24	33	9	0	0	0	0
	CDH26.7	8	0	0	0	0	0	2	6	1	0	0	0	0
Wind	WSAvg	2.4	2.3	2.8	2.7	2.8	2.6	2.1	1.9	1.9	2.2	2.6	2.4	2.2
Precipitation	PrecAvg	604	40	38	33	28	36	47	67	85	75	56	52	46
	PrecMax	782	103	88	102	54	77	78	117	126	135	120	123	95
	PrecMin	460	4	5	3	6	10	23	29	37	15	7	4	14
	PrecStd	100	22	20	22	13	14	12	19	25	30	25	29	20
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.4	6.0	6.2	13.8	22.8	26.3	27.2	24.1	17.9	12.8	7.6	7.0
		MCWB	2.8	2.6	2.5	6.2	12.4	15.3	17.1	16.7	11.9	8.8	4.9	3.2
	2%	DB	4.1	4.0	4.0	11.3	19.1	23.0	23.1	20.7	16.1	11.0	5.9	4.1
		MCWB	1.5	1.5	1.0	5.1	10.6	13.9	15.3	15.1	11.0	7.6	3.6	1.7
	5%	DB	2.9	2.9	2.8	8.9	17.1	21.2	21.5	18.0	14.0	8.8	3.6	2.6
		MCWB	0.7	0.6	0.4	4.3	9.7	13.3	14.8	13.4	10.1	6.3	1.9	0.7
	10%	DB	1.9	2.3	1.8	7.3	14.1	18.8	19.1	17.1	12.7	7.4	2.2	1.0
		MCWB	0.2	0.3	-0.2	3.3	8.0	12.4	14.0	13.0	9.7	5.2	0.8	-0.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.5	3.0	3.0	6.7	13.2	16.8	18.7	17.5	12.9	9.5	5.3	4.2
		MCDB	6.3	5.0	5.7	12.4	21.1	23.9	25.5	22.8	16.5	12.7	7.3	6.6
	2%	WB	2.0	1.8	1.6	5.4	11.5	14.9	16.2	15.5	11.7	7.8	3.7	2.0
		MCDB	4.1	3.9	3.7	10.6	18.3	21.6	22.1	19.0	14.8	10.0	5.6	4.1
	5%	WB	1.1	1.1	0.7	4.4	10.0	13.8	15.2	14.3	10.8	6.6	2.0	0.9
		MCDB	3.2	2.9	2.8	8.6	16.1	19.7	20.3	17.5	13.3	8.6	3.5	2.8
	10%	WB	0.0	0.3	-0.4	3.4	8.7	12.7	14.3	13.3	10.1	5.2	0.6	-0.4
		MCDB	1.9	2.0	1.5	7.1	13.9	18.0	18.8	16.4	12.5	7.4	1.9	1.1
Mean Daily Temperature Range	5% DB	MDBR	7.1	8.1	9.4	8.8	9.4	9.7	8.0	7.2	7.2	5.9	6.6	7.4
		MCDBR	6.6	6.1	7.8	11.3	13.7	14.4	12.6	10.5	9.4	6.2	5.7	6.5
		MCWBR	5.4	4.9	5.9	6.8	7.3	7.0	6.3	6.1	5.8	4.2	4.7	5.3
	5% WB	MCDBR	6.6	6.1	7.1	10.1	12.4	12.4	11.2	8.9	8.2	5.8	5.5	6.3
		MCWBR	5.4	5.1	5.5	6.2	6.7	6.4	5.9	5.3	5.3	4.2	4.7	5.3
Clear-Sky Solar Irradiance	taub	0.222	0.270	0.279	0.344	0.370	0.365	0.363	0.370	0.320	0.290	0.230	N/A	
	taud	2.057	2.208	2.256	2.167	2.311	2.382	2.429	2.411	2.531	2.419	2.071	N/A	
	Ebn at Noon	492	707	844	839	849	861	853	812	804	690	463	N/A	
	Edh at Noon	37	65	88	119	113	107	100	94	69	53	35	N/A	
All-Sky Solar Radiation	RadAvg	0.33	1.02	2.89	4.55	5.08	5.23	4.38	3.34	2.31	1.09	0.40	0.16	
	RadStd	0.05	0.12	0.34	0.43	0.52	0.45	0.41	0.34	0.30	0.10	0.06	0.02	